

# Skills Summary

## Reading and Making Line Plots

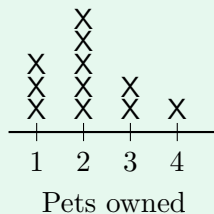
Use this reference while completing your worksheet or when studying.

### Skill 1 — Reading how many from a stack of X's

**A line plot holds two kinds of number.** The numbers written *along the bottom* are the thing that was counted or measured. The X's stacked *above* a number say how many times that thing happened. One X is always one thing.

So “how many” questions are answered by counting a stack, never by reading a label. A number with an empty space above it is not a mistake: it means nothing landed there.

**Example:** How many children own 2 pets?



**Step 1.** The question asks how many *children*, so the answer is the size of a stack — find the 2 along the bottom and count upwards, do not read the label.

**Step 2.** Above the 2: X, X, X, X, X.

⇒ 5 children own 2 pets.

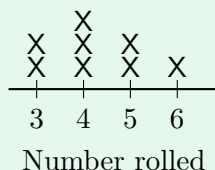
**Try it:** On the same plot, how many children own 3 pets?

check: 2 children

## Skill 2 — Making a line plot from a list

**Three steps.** (1) Find the smallest and largest number on the list, and write every whole number from one to the other along a line — in order, evenly spaced, *including the ones that never appear*. (2) Go along the list and draw one X above each number, crossing it off the list as you go. (3) Count the X's: there must be exactly as many as there were numbers on the list.

**Example:** Plot 3, 4, 5, 4, 3, 4, 6, 5. How many X's sit above 4?



**Step 1.** The list runs from 3 to 6, so the line needs 3, 4, 5, 6 — then one X per number on the list, crossing each off as it is drawn.

**Step 2.** Crossing off the 4's one at a time gives three of them, and the eight X's on the plot match the eight numbers on the list.

⇒ **3 X's** sit above the 4.

**Try it:** On the same plot, how many X's sit above 5?

check: 2 X's

## Skill 3 — Totalling and comparing stacks

**How many altogether?** Add the sizes of all the stacks — that is the total number of X's, not the number of marks along the bottom.

**How many more?** Subtract one stack size from the other.

**How much bigger / longer?** That answer is about the thing measured, so subtract the two *labels* from the bottom of the line instead.

**Example:** On the pets plot (3, 5, 2 and 1 X's above 1, 2, 3 and 4), how many children were asked altogether?

**Step 1.** Every child put exactly one X on the plot, so counting the children means adding up all four stacks.

**Step 2.** Add the four stack sizes:  $3 + 5 + 2 + 1$ .

⇒ **11 children** were asked.

**Try it:** On the same pets plot, how many more children own 2 pets than own 3 pets?

check: 3 more children

**(!) Watch out:** the commonest line-plot mistake is answering with a label instead of a stack. Before writing anything down, ask yourself: *is this answer counting things, or measuring them?* Counting means a stack of X's; measuring means a number from the bottom of the line.